

Ficks law

$$J_x = -D \cdot \frac{dc}{dx}$$

flux positive in x direction

units

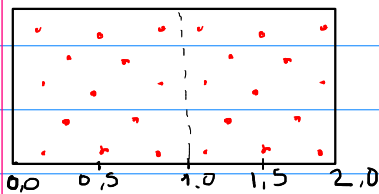
$$J: \frac{\text{moles}}{\text{m}^2 \cdot \text{s}}$$

$$D: \frac{\text{m}^2}{\text{s}}$$

$$\frac{dc}{dx}: \frac{\text{moles}}{\text{m}^3 \cdot \text{m}} = \frac{\text{moles}}{\text{m}^4}$$

Ficks first law: how flux depend only on concentration gradient

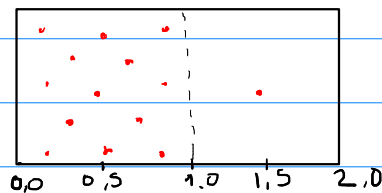
Lumped model



$$\text{flux inside system} \\ = -D \cdot \frac{dc}{dx} = -D \cdot 0 = 0$$

distributed model

time = 0

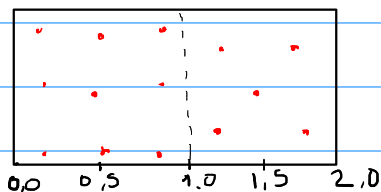


$$\text{flux} = -D \cdot \frac{dc}{dx} = -D \cdot \frac{1-13}{1,5-0,5} = -D \cdot \left(\frac{-12}{1}\right) = D \cdot 12$$

big flux

distributed model

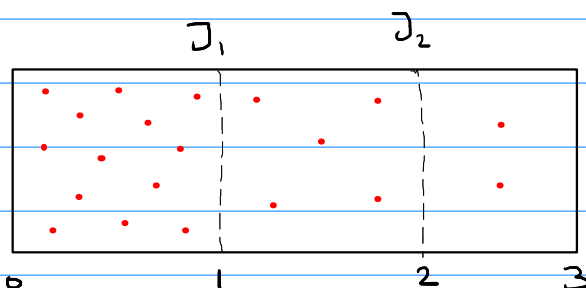
time = 10



$$\text{flux} = -D \cdot \frac{dc}{dx} = -D \cdot \frac{5-9}{1,5-0,5} = -D \cdot \left(\frac{-4}{1}\right) = D \cdot 4$$

lesser flux

flux change with time and space!



$$\frac{dJ}{dx} = \frac{J_2 - J_1}{2 - 1} = \frac{\frac{\text{moles}}{\text{m}^2 \cdot \text{s}}}{\text{m}} = \frac{-\text{moles}}{\text{m}^3 \cdot \text{s}} = -\frac{dc}{dt}$$

$$\frac{dC}{dt} = -\frac{dJ}{dx} \quad \frac{dC}{dt} = -\frac{d\left(-D\frac{dC}{dx}\right)}{dx} = D \cdot \frac{d^2C}{dx^2}$$

substitute first law

D is constant

Fick's second law

$$\frac{dC}{dt} = D \cdot \frac{d^2C}{dx^2}$$